

Raymond Li

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EXPERIENCE

Meta

Menlo Park, CA

Software Engineer, AI Infra - Experiences ML

2025 – Present

- Building a training and inference [analysis platform](#) for GPU utilization and kernel optimization
- Re-architected monolithic analysis pipeline into highly-parallel Prefect-like DAG workflow
 - Reduced latency by **70%** (**7000 hours** of runtime saved **daily** across **150k** runs)
 - Eliminated OOMs, boosting overall execution success rate from **82%** to **99%**
 - Enabled partial retries on failures and improved observability (critical for SEV root-causing)
- Introducing Protobuf to [PyTorch/Kineto](#) for **3x** speedup and **85%** less memory than JSON traces
- **100%** AI-native development since Oct 2025. Built Claude skills to solve domain-specific and general human-on-the-loop challenges, replacing manual work in various internal processes.

Production Engineer, PyTorch

2024 – 2025

- Drove **4%** speedup in `torch.compile`, saving **\$10M** annual recurring infra costs, via optimization of node fusion in `TorchInductor`. Also improved PT2 compile observability, reliability and scalability.
- Built `torch.compile` efficiency analysis system and gleaned insights for top-10 models company-wide, providing data for org-level PyTorch compile engineering decision-making

Tesla

Palo Alto, CA

Software Engineering

2024

- Built multi-layered caching API for `SCons` build system to speed up builds by **70%** (Python, Linux)
- Automated tedious manual build-test-deploy process through a `Jenkins` CI pipeline and `Docker` containers, saving **3 days** per release cycle for core library used to build **200+** components
- Spearheaded upgrade of all (**9**) components built on `Ubuntu` 12.04 to 20.04. Full toolchain upgrades (e.g. `GCC` 4.X -> 10.5). Coordinated **16** Engineers in **5+** teams across **3** orgs.

Amazon Web Services (AWS)

Seattle, WA

Software Dev Engineering

2023

- Developed `Python` job to clean up **30K+** incommunicado (**96%** of all) `DynamoDB` Transaction-Coordinator nodes, significantly reducing `EC2` usage costs and receiving SVP commendation

CharaChorder

Dallas, TX

Open Source

2023 –

- Led development of [nexus](#), a cross-platform typing analysis desktop app, and CharaChorder Zero, a software-emulated version of fluid chorded-character entry hardware devices

PlayStation (Sony Interactive Entertainment)

Waterloo, ON

Software Dev Engineering

2022

- Independently conceived, designed, developed and documented **first ever PS5 Easter Egg**
- Migrated **50%** of custom `React Native` tile components in PS5 Control Center to SDK, simplifying codebase, reducing compilation time and improving maintainability

Software Dev Engineering in Test

2021

- First prize, PlayStation worldwide hackathon (game-recommendation app, out of **100+** teams)
- Developed Chrome Extension to compare different versions of test specifications, and scripts to automatically sort, version, label and template `pytest` test scripts, reducing time required by 90%

PROJECTS

[i3lock-color](#)  C, X11

1K+ stars, 130+ forks, 10K+ users

- World's most popular non-default computer lockscreen

[VoiceBox](#)  React.js, GCP

- Text-to-speech webapp using Google Cloud AI, acquired by non-profit education org

EDUCATION

University of Waterloo

Bachelor of Computer Science, Dean's Honours

- 3.9 GPA. Co-designed [CS 136L](#) Linux course (incl. Bash, Git, I/O Testing, Make, Debugging, IDEs)

UW Computer Science Club Sysadmin

- Led two teams (Systems Committee and Terminal Committee) of **20+** CS students to configure, upgrade and maintain **35+** servers, on-prem web- and email-hosting, LDAP, Gitea, IRC/ZNC, NextCloud, video-conferencing (BBB), MediaWiki, CloudStack, and K8s
- Responsible for the largest open-source software mirror in Canada
 - **100k+** HTTP requests/hour, **1.6 petabytes** (1600 TB) of data served annually

Undergraduate Research Assistant

- Built various ML models (KNN, SVM, random forest, etc) using **Python**, **pandas**, **NumPy** and **scikit-learn** to classify pilots' experience levels based on ALSIM simulator data of real pilot landings, creating world's-first system for AI-driven pilot experience evaluation.

SKILLS

Languages: Python, JavaScript (React, Node), Kotlin/Java, C/C++, HTML/CSS, Bash

Technologies: Git, Docker, Jenkins, GCP/AWS/Cloudflare, Apache, Arch Linux, WSL

AI Systems: Claude code, OpenAI Codex, Gemini, OpenCode, OpenClaw, multi-agent systems

Interests: Enjoys snowsports, horse riding, flying (GA), top-roping, chess. Plays violin. #btwluseArch